

Amanda Li

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EDUCATION

The Bear Creek School (Class of 2027)

Redmond, WA

August 2023 — June 2027

- Cumulative GPA: 4.0 (unweighted)
- Relevant Coursework: AP Computer Science A (5), AP Calculus AB (5), AP Chemistry (5), AP Physics 1 (5)

RESEARCH AND WORK EXPERIENCE

Researcher for Space Science at UT Arlington

April 10th Grade — Present

- Implemented machine-learning denoising pipelines (Random Forest, XGBoost, LSTM) under Professor Zhonghua Xu to reduce temperature-correlated noise by **89.179%** in NASA EZIE low-cost magnetometer data, addressing a core limitation in scientific usability of low-cost geomagnetic sensors
- Built a fully automated end-to-end processing pipeline handling 170,000+ time-series samples (1-second resolution), including despiking, time alignment with FRD IAGA-2002 observatory data, lag feature generation, and regression
- Trained and optimized XGBoost models achieving test RMSE as low as 12.44 nT and R^2 up to 0.89, enabling high-fidelity recovery of true geomagnetic field signals

Researcher for Cybersecurity at the University of Washington

March 10th Grade — Present

- Conducting cybersecurity research under Professor Marc Dupuis on phishing susceptibility, social engineering psychology, and memory decay in security training effectiveness involving 100+ participant
- Manually curated, filtered, and labeled a high-fidelity dataset of 200+ emails (100+ phishing, 100+ legitimate) to support behavioral and statistical modeling
- Leading the design and execution of a human-subjects experiment involving 100+ participants to measure phishing detection accuracy, training retention, and long-term memory decay

Paid Internship at SayWeee

July 10th Grade — August 10th Grade

- Worked on inventory tracking and supply-chain optimization across 9 warehouse facilities and 1,000+ SKUs, addressing out-of-stock failures in the SayWeee online shopping platform
- Analyzed order flow, stocking delays, and warehouse throughput bottlenecks to identify sources of fulfillment breakdown
- Assisted in workflow restructuring and inventory data organization, contributing to reduced product unavailability and improved order completion reliability
- Collaborated with operations, logistics, and engineering teams in a high-volume, real-time e-commerce environment

PERSONAL PROJECTS

CipherCats

June 10th Grade — Present

- Built a full-stack cryptography learning platform with 10 interactive Linux-based labs teaching RSA, AES, block ciphers, hashing, and modular arithmetic
- Engineered containerized execution environments and scalable React + Flask + Supabase architecture for secure execution, authentication, and progress tracking

ChonkOS II - An Operating System Built from Scratch

February 10th Grade — March 10th Grade

- Built a fully functional x86 operating system from scratch using C and x86 Assembly, targeting Intel 8086 backward compatibility
- Implemented core OS subsystems including a custom bootloader, FAT16 filesystem, protected-mode transitions, interrupt handling, and low-level memory management
- Developed bare-metal execution environment inside QEMU, enabling complete system debugging and introspection
- Gained deep understanding of hardware-software interaction, firmware security, system call mechanics, and low-level exploit surfaces

EXTRACURRICULAR

Founder & President – CyberTalks (501(c)(3) Nonprofit)

September 10th Grade — Present

- Founded a cybersecurity education nonprofit delivering digital safety training to senior communities
- Designed curriculum on phishing detection, scam prevention, password security, and online identity protection
- Delivered in-person workshops to 50+ participants, directly addressing high-risk vulnerability to cyber fraud among elderly populations

MIT Beaverworks Summer Institute (BWSI) Program

July 9th Grade — August 9th Grade

- Attended the Embedded Security and Hardware Hacking Program
- Designed and implemented a secure firmware transmission system, addressing firmware tampering and interception risks
- Worked extensively with Ghidra, GDB, UART interfaces, and firmware reverse engineering
- Awarded Dr. Bob Berman Award for Disruptive Engineering (1st out of 32 students)

VEX VRC Robotics Team Lead

August 8th Grade — Present

- Led Team 10W as project manager and lead programmer, engineering autonomous navigation and real-time control systems (PID + odometry)
- Designed PID control loops and odometry-based navigation systems, enabling high-precision autonomous movement
- Achieved Division Champion + Build Award at 2025 Worlds, Teamwork Award at 2024 Worlds, Top 14 globally out of 30,000+ teams, and 2× world records
- Set two world records for highest match scoring
- Featured on Pits and Parts, a leading international robotics media channel

Cybersecurity YouTube Channel

August 8th Grade — Present

- Founded and manage a cybersecurity education channel explaining security vulnerabilities, attacks, and digital safety concepts
- Independently write scripts, design visuals, and edit videos using Premiere Pro, After Effects, and Photoshop
- Achieved 10,000+ total views, demonstrating high-impact technical communication and educational outreach

Stanford Intro to Logic Summer Program

June 10th Grade — July 10th Grade

- Completed the Stanford Introduction to Logic program under Professor Michael Genesereth, studying formal reasoning through propositional, modal, and epistemic logic
- Gained proficiency in constructing logical proofs, applying inference rules (e.g., implication elimination), and clearly articulating structured, symbolic arguments

CERTIFICATES AND SKILLS

- **2026:** CompTia Security+
- **2025:** CompTia A+
- **2024:** MIT BWSI Embedded Security and Hardware Hacking Certificate || SANS GIAC Foundational Cybersecurity Technologies Certificate || Texas A&M GenCyber Certificate
- **Security Skills:** Embedded exploitation, firmware analysis, reverse engineering, web exploitation, cryptography
- **Programming Languages:** C/C++, Python, Java, JavaScript, Bash, x86 ASM, R, React, Typescript
- **Systems & Tools:** Linux (Kali, Ubuntu), Docker, Git, GDB, QEMU, Wireshark, BurpSuite, Nmap, SSH, Ghidra, Supabase, Vercel

AWARDS

- 2026: Vex Division Champion (World Qualifying)
- 2025: VEX World Championship Division Champion || Build Award (World Champion and State Level) || 2× VEX World Record Holder || CIE-USA Scholarship Award
- 2024: MIT BWSI Dr. Bob Berman Award || National Cyber Scholar (\$1200 scholarship) || VEX Worlds Outstanding Teamwork Award || CIE-USA Special Award